Мышцы ягодичной области

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Мышцы ягодичной области — версия для подкаста
TeachMeAnatomy

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Ягодичная область — это анатомическая область, расположенная кзади от тазового пояса, в проксимальном отделе бедренной кости. Мышцы этой области обеспечивают движение нижней конечности в тазобедренном суставе.

Мышцы ягодичной области можно условно разделить на две группы:

- **Поверхностные отводящие и разгибающие мышцы** — группа крупных мышц, отводящих и разгибающих бедро. К ним относятся большая ягодичная мышца, средняя ягодичная мышца, малая ягодичная мышца и напрягатель широкой фасции бедра.
- **Глубокие мышцы, вращающие бедро в стороны** — группа более мелких мышц, которые в основном отвечают за вращение бедра в стороны. К ним относятся квадратная мышца бедра, грушевидная мышца, верхняя близнецовая мышца, нижняя близнецовая мышца и внутренняя запирательная мышца.

Артериальное кровоснабжение этих мышц осуществляется в основном за счет верхней и нижней ягодичных артерий — ветвей **внутренней подвздошной артерии**. Венозный отток происходит по ходу артерий.

В этой статье мы рассмотрим две группы ягодичных мышц — их прикрепление, иннервацию и функции. Мы также расскажем о клинических проявлениях нарушений в работе ягодичных мышц.

Поверхностные мышцы

Поверхностные мышцы ягодичной области состоят из трех ягодичных мышц и широкой фасции бедра. В основном они отвечают за отведение и разгибание нижней конечности в тазобедренном суставе.

Большая ягодичная мышца

Большая ягодичная мышца — самая крупная из ягодичных мышц. Она также является самой поверхностной и формирует форму ягодиц.

- **Прикрепление**: начинается от ягодичной (задней) поверхности подвздошной кости, крестца и копчика. Волокна пересекают ягодицу под углом 45 градусов и прикрепляются к подвздошно-большеберцовому тракту и ягодичному бугорку бедренной кости.
- **Функции**: является основным разгибателем бедра и участвует в наружной ротации. Однако задействуется только при необходимости приложить силу, например при беге или подъеме по лестнице.
- **Иннервация**: нижний ягодичный нерв.

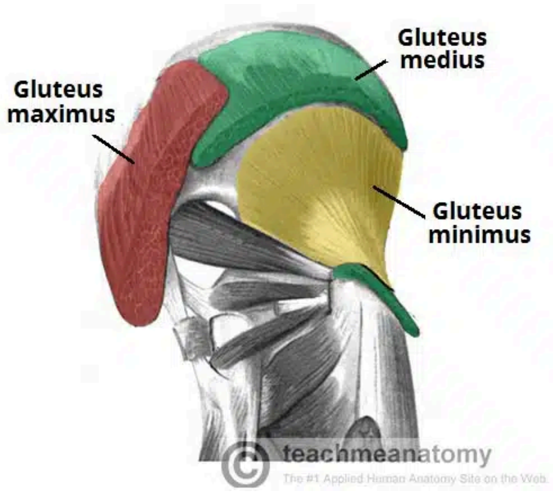

Fig 1

The superficial muscles of the gluteal region. The gluteus maximus and medius have been partly removed.

Gluteus Medius

The gluteus medius muscle is fan-shaped and lies between to the gluteus maximus and the minimus. It is similar in shape and function to the gluteus minimus.

- **Attachments**: Originates from the gluteal surface of the ilium and inserts into the lateral surface of the greater trochanter.
- **Actions**: Abduction and medial rotation of the lower limb. It stabilises the pelvis during locomotion, preventing 'dropping' of the pelvis on the contralateral side.
- **Innervation**: Superior gluteal nerve.

Gluteus Minimus

The gluteus minimus is the deepest and smallest of the superficial gluteal muscles. It is similar in shape and function to the gluteus medius.

- **Attachments**: Originates from the ilium and converges to form a tendon, inserting to the anterior side of the greater trochanter.
- **Actions**: Abduction and medial rotation of the lower limb. It stabilises the pelvis during locomotion, preventing 'dropping' of the pelvis on the contralateral side.

- **Innervation**: Superior gluteal nerve.

Tensor Fascia Lata

Tensor fasciae lata is a small superficial muscle which lies towards the anterior edge of the iliac crest. It functions to tighten the fascia lata, and so abducts and medially rotates the lower limb.

- **Attachments:** Originates from the anterior iliac crest, attaching to the anterior superior iliac spine (ASIS). It inserts into the iliotibial tract, which itself attaches to the lateral condyle of the tibia.
- **Actions:** Assists the gluteus medius and minimus in abduction and medial rotation of the lower limb. It also plays a supportive role in the gait cycle.
- **Innervation:** Superior gluteal nerve.

CLINICAL RELEVANCE

Damage to the Superior Gluteal Nerve

The **superior gluteal nerve** innervates the gluteus medius and the gluteus minimus. These muscles have an important role in stabilising the pelvis during locomotion. In the standing position, the gluteus minimus and medius contract when the contralateral leg is raised, preventing the pelvis from dropping on that side.

If the superior gluteal nerve is damaged, the previously described muscles are paralysed – and the pelvis becomes unsteady. A characteristic finding of gluteal muscle weakness is the **Trendelenburg sign**.

Trendelenburg Sign

The Trendelenburg sign is produced when the patient is asked to stand unassisted on each leg in turn. In a positive sign, **pelvic drop** will occur on the unsupported leg. Pelvic drop can be recognised by observing the level of the **iliac crests** on both sides.

For example, if the left gluteal muscles are weak, the right side of the pelvis will drop when the patient stands on their left leg (and the right leg is unsupported).

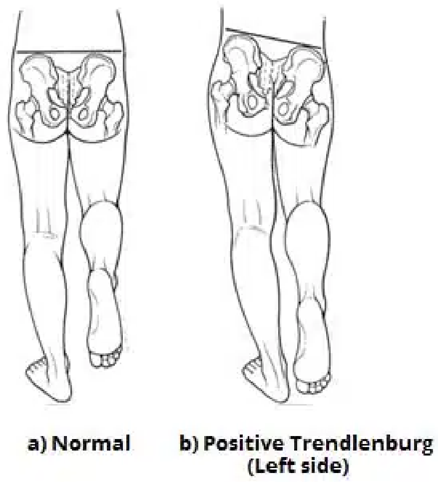

Fig 2

Positive Trendelenburg sign, characteristic of left superior gluteal nerve palsy.

The Deep Muscles

The deep gluteal muscles are a set of smaller muscles, located underneath the gluteus minimus. The general action of these muscles is to laterally rotate the lower limb. They also stabilise the hip joint by ‘pulling’ the femoral head into the acetabulum of the pelvis.

<u>Piriformis</u>

The piriformis muscle is a key landmark in the gluteal region. It is the most superior of the deep muscles.

- **Attachments**: Originates from the anterior surface of the sacrum. The fibres travel inferiorly and laterally through the greater sciatic foramen to insert onto the greater trochanter of the femur.
- **Actions**: Lateral rotation and abduction.

- **Innervation**: Nerve to piriformis.

Obturator Internus

The obturator internus forms the lateral walls of the pelvic cavity. In some texts, the obturator internus and the gemelli muscles are considered as one muscle – the triceps coxae.

- **Attachments**: Originates from the pubis and ischium at the obturator foramen. It travels through the lesser sciatic foramen, and attaches to the greater trochanter of the femur.
- **Actions**: Lateral rotation and abduction.
- **Innervation**: Nerve to obturator internus.

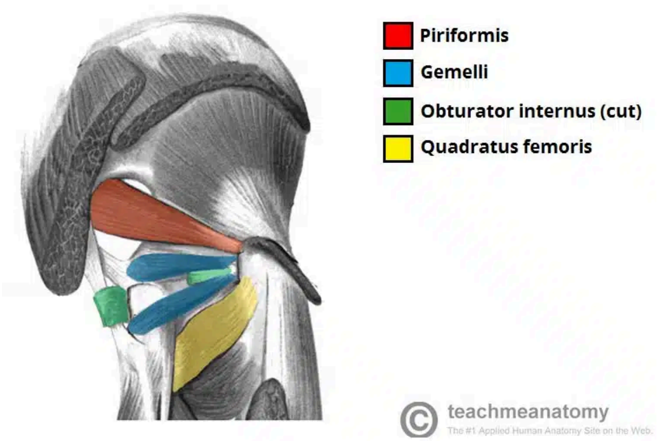

Fig 3

The deep muscles of the gluteal region.

The Gemelli – Superior and Inferior

The gemelli are two narrow and triangular muscles. They are separated by the obturator internus tendon.

- **Attachments**: The superior gemellus muscle originates from the ischial spine, the inferior from the ischial tuberosity. They both attach to the greater trochanter of the femur.
- **Actions**: Lateral rotation and abduction.
- **Innervation**: The superior gemellus muscle is innervated by the nerve to obturator internus, the inferior gemellus is innervated by the nerve to

quadratus femoris.

Quadratus Femoris

The quadratus femoris is a flat, square-shaped muscle. It is the most inferior of the deep gluteal muscles, located below the gemelli and obturator internus.

- **Attachments**: Originates from the lateral aspect of the ischial tuberosity and attaches to the quadrate tuberosity on the intertrochanteric crest.
- **Actions**: Lateral rotation.
- **Innervation**: Nerve to quadratus femoris.

CLINICAL RELEVANCE

Landmark of the Gluteal Region

The piriformis is an important **anatomical landmark** in the gluteal region.

As the muscle travels through the greater sciatic foramen, it effectively divides the gluteal region into an inferior and superior part. This division determines the name of the vessels and nerves that supply the area. The **superior gluteal nerve** and vessels emerge into the gluteal region superiorly to the piriformis (and vice versa for the inferior gluteal nerve).

In addition, the piriformis can be used to locate the **sciatic nerve** (a major peripheral nerve of the lower limb). The sciatic nerve enters the gluteal region directly inferior to the piriformis, and is visible as a flat band, approximately 2cm wide.

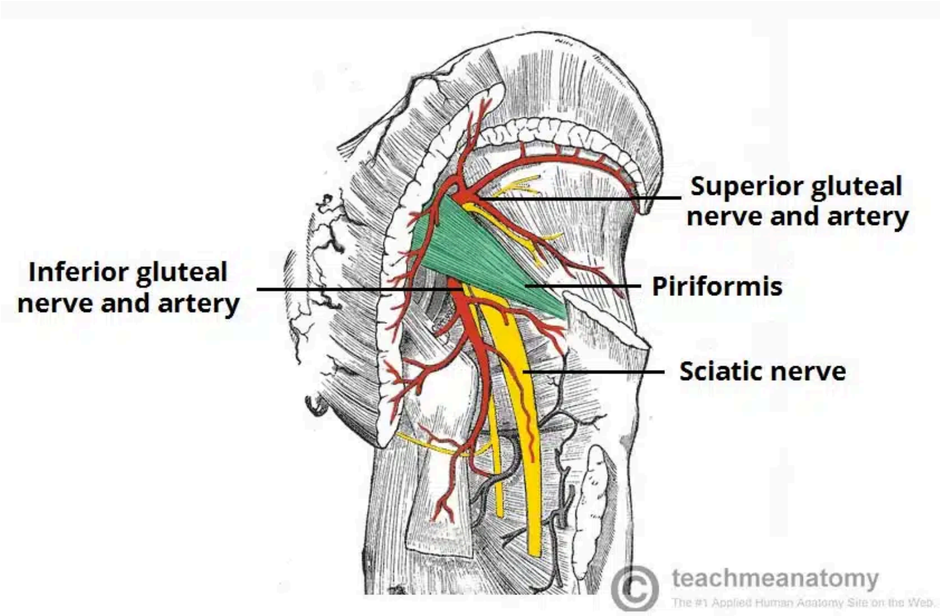

Fig 4

The piriformis as an anatomical landmark in the gluteal region.